PRODUCTION PLANNING

Production Planning can be referred to as a technique of forecasting every step in the long process of production, taking them at right time and in the right degree and trying to complete operations at the maximum efficiency.

"The planning of industrial operations involves Three considerations, namely, what work shall be done, how the work shall be done and lastly, when the work shall be done." (by – Kimball)

PRODUCTION CONTROL

Production control is the process that keeps a watchful eye on the production flow, size of resources along with any deviation from the planned action. It also includes arrangement for the prompt remedy or adjustment in case of any deviation so that the production may run according to the original or revised schedule.

"Production control refers to ensuring that all which occurs is in accordance with the rules established and instructions issued." (By – Henry Fayol)

PPC : HISTORY

- **1776** -Specialization of labor in manufacturing -**Adam Smith**
- **1799** -Interchangeable parts, cost accounting **-Eli Viihitney and others**
- **1832** -Division of labor by skill; assignment of jobs by skill; basics of time study -**Charles Babbage**
- **1900**- Scientific management time study and work study developed; dividing planning and doing of work -**Frederick W. Taylor**
- **1900**- Motion of study of jobs -**Frank B. Gilbreth**
- **1901**- Scheduling techniques for employees, machines jobs in manufacturing -**Henry L. Gantt**
- **1915** -Economic lot sizes for inventory control -**F.W.**
- **Harris**

PPC : HISTORY

- **1927** -<u>Human relations; the Hawthorne studies</u> **-Elton Mayo**
- **1931** -Statistical inference applied to product quality: quality control charts -**W.A. Shewart**
- **1935** -Statistical sampling applied to quality control; inspection sampling plans -**H.F. Dodge &H.G. Roming**
- **1940**- Operations research applications in World War ll -**P.M. Blacker and others.**
- **1946**- Digital computer -**John Mauchlly and J.P. Eckert**
- **1947**-Linear programming -**GB. Dantzig, Williams & others**

PPC : HISTORY

- **1950**- Mathematical programming, on-linear and stochastic processes –**A. Charnes, W.W. Cooper & others**
- **1951**- Commercial digital computer; large scale computations available. -**Sperry Univac**
- **1960**- Organizational behavior; continued study of people at work -**L. Cummings, L. Porter**
- **1970**- Integrating operations into overall strategy and policy. Computer applications to manufacturing. Scheduling and control. Material requirement planning (MRP)-**W. Skinner J. Orlicky and G. Wright**
- **1980**-Quality and productivity applications from Japan robotics. CAD-CAM -**W.E. Deming and J. Juran**

PPC : HISTORY

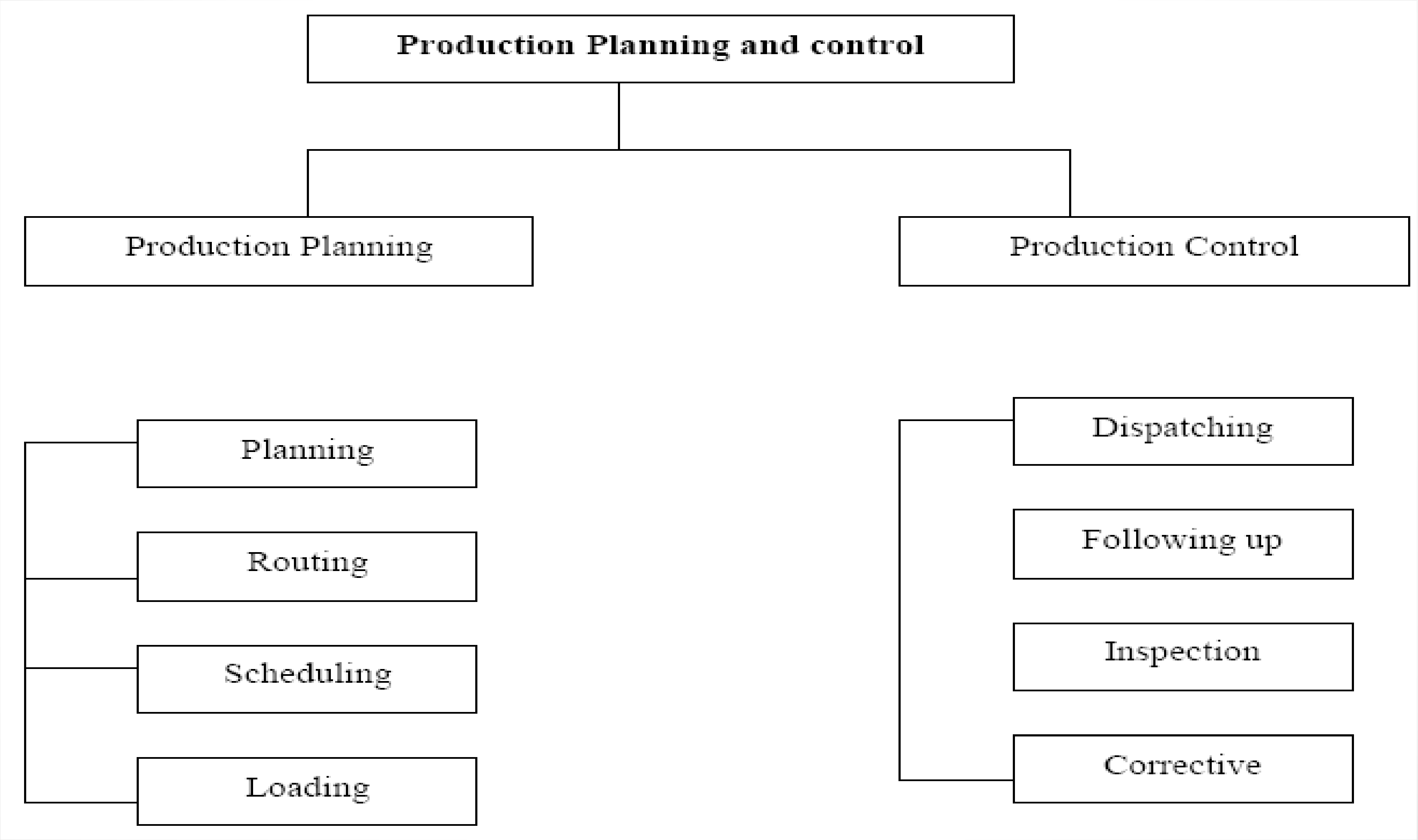
- PPC concept developed since late 19th Century
- Factories were simple and relatively small
- Small number of products with large batches
- Work for each man and each machine used to be chalked out
- even as factories grew,they were just bigger,not more complex
- Main Industry – Textiles , Railways

PPC : OBJECTIVES

PRODUCTION PLANNING involves….

- Preparation of production budget
- Devising manufacturing methods and sequence of operations
- Deciding type of machines and equipments
- Preparation of operation sheets and instruction cards
- Estimating men, machine and material requirements
- Undertaking time and motion stu
- Preparing master schedules

PPC : PROCESS



Production Planning and control
Production Planning
Production Control
Planning
Routing
Scheduling
Loading
Dispatching
Following up
Inspection
Corrective

PLANNING

It is the first element of production planning and control. Planning means deciding in advance what is to be done in future. An organizational set up is created to prepare plans and policies. Various charts, manuals and production budgets are also prepared. Planning provides a sound base for control. A separate department is set up for this work.

ROUTING

Routing is determining the exact path which will be followed in production. It is the selection of the path from where each unit have to pass before reaching the final stage. The stages from which goods are to pass are decided in this process.

“Routing is the specification of the flow sequence of operations and processes to be followed in producing a particular manufacturing lot.”(Alford & Beaty)

ROUTING PROCEDURE

- Deciding what part to be made or purchased
- Determining Materials required
- Determining Manufacturing Operations and Sequences
- Determining of Lot Sizes
- Determining of Scrap Factors
- Analysis of Cost of the Product
- Preparation of Production Control Forms

SCHEDULING

Scheduling is the determining of time and date when each operation is to be commenced or completed. The time and date of manufacturing each component is fixed in such a way that assembling for final product is not delayed in any way.

"The determination of the time that should be required to perform each operation and also the time necessary to perform the entire series, as routed, making allowances for all factors concerned."(Kimball)

TYPES OF SCHEDULES

- **Master Scheduling** – It is the breakup of production requirements. It is the start of scheduling. It is prepared by keeping in view the order or likely sales order in near future.
- **Manufacturing Scheduling –** It is used where production process is continuous. The order of preference for manufacture is also mentioned in the schedule for a systematic production planning.
- **Detail Operation Scheduling** – It indicates the time required to perform each and every detailed operations of a given process

LOADING

- The next step is Loading which is execution of the scheduled plan as per the route chalked out. It includes the assignment of the work to the operators at their machines or work places.
- So Loading determines who will do the work.

DISPATCHING

Dispatching refers to the process of actually ordering the work to be done. It involves putting the plan into effect by issuing orders. It is concerned with starting the process and operation on the basis of route sheets and schedule charts.

“Dispatches put production in effect by releasing and guiding manufacturing order in the sequence previously determined by route sheets and schedules.”(John A.Shubin)

DISPATCHING PROCEDURES

- **Centralized Dispatching –** Under this, orders are directly issued to workmen and machines. It helps in exercising effective control.
- **Decentralized Dispatching –** Under this procedure all work orders are issued to the foreman or dispatch clerk of the department or section. It suffers from difficulties in achieving co-ordination among different departments.

FOLLOW UP & EXPEDITING

Progress may be assessed with the help of routine reports or communication with operating departments. The follow up procedure is used for expediting and checking the progress.

"Follow up or expediting is that branch of production control procedure which regulates the progress of materials and part through the production process."

INSPECTION

- Inspection is the process of ensuring whether the products manufactured are of requisite quality or not.
- Inspection is undertaken both of products and inputs. It is carried on at various levels of production process so that pre-determined standards of quality are achieved.
- Inspection ensures the maintenance of pre-determined quality of products.

CORRECTIVE MEASURES :

- Adjusting the route
- Rescheduling of work
- Changing the workloads
- Repairs and Maintenance of machinery or equipment,
- Control over inventories
- Certain personnel decisions like training, transfer, demotion etc.
- Alternate methods may be suggested to handle peak loads.

PPC : LIMITATIONS

- Assumption based
- Rigidity
- Difficult for small firms
- Costly
- Dependence on external factors
- Team work is a must
- Demands high level of co-ordination & efficiency

PPC : SIGNIFICANCE

- Structured & Planned Process
- Increased Production
- Seamless Plant Activity
- Better Co-ordination
- Optimal Resource Utilization
- Cost Control
- Rationalization of production Activities